CHAPTER

26

Tooth Substance Loss

With the dramatic evolution of dental materials coupled with patient's awareness regarding preventive regimes, the individuals retaining their original dentition is increasing day by day. With the retention of natural dentition, the management of loss of tooth substance is posing a challenge for the clinicians.

Apart from caries, non-carious processes, such as attrition, abrasion and erosion may cause loss of tooth structure. The etiology varies among different individuals; may be associated with physiological or pathological processes.

Eccles suggested the term '*tooth substance loss*', which can be used to encompass terms, such as attrition, abrasion and erosion. Later, Smith and Knight (1984) proposed the term 'tooth wear' so as to include all these conditions and their combination. They contended that the term *'tooth surface loss'* did not sufficiently reflect the severity of the condition and advocated use of the term 'tooth wear'. They also proposed a tooth wear index as depicted in Table 26.1. Later, Bardsley et al (2004) simplified the tooth wear index for epidemiological studies (Table 26.2).

The types of wear can be as follows:

ATTRITION

This term is derived from Latin word '*attitum*', which describes the action of rubbing against any object.

Dental attrition is defined as "*the physiologic wearing of teeth resulting from tooth-to-tooth contact as in mastication. This is an age-related process that can occur at the incisal or occlusal surfaces and sometimes on proximal surfaces*".

Clinically, the first manifestation is the appearance of a small polished facet on a cusp tip or ridge or an incisal edge. Severe attrition, however, can result in dentinal exposure, which may increase the rate of wear (Fig. 26.1A, B).

Attrition may be hastened by a hard and abrasive diet or inorganic (silica compound) particles in allied food items (tobacco, etc.). However, a few authors disagree with the food as a major factor. They prefer to use the term '*demastication*' to describe the wearing away of tooth substance during mastication influenced by the abrasiveness of the individual food particles.

Some parafunctional habits, like bruxism and clenching, may also contribute to attrition. A good number of individuals may be affected by such habits. A certain amount of attrition is considered physiologic; however, excessive destruction of tooth structure is pathological. If occlusal wear occurs at a rate faster than the compensatory physiologic mechanisms, it is not 'physiological' (Fig. 26.2).

At times, severe occlusal wear causes amalgam/any restoration to appear protruded, while the surrounded occlusal surface is flattened (Fig. 26.3).

Table 26.1: Tooth wear index

Score	*Buccal/lingual*	*Interproximal/occlusal*	*Cervical*
0	No loss of enamel surface characteristics	No loss of enamel surface characteristics	No change in contour
1	Change in enamel surface characteristics	Change in enamel surface characteristics	Minimal loss of contour
2	Loss of 1/3rd of enamel surface	Loss of enamel fissure exposing dentin	Defect <1.0 mm
3	Loss of more than 1/3rd of enamel surface	Exposure of dentin	1.0–2.0 mm deep defect
4	Complete loss of enamel surface; exposure of dentin	Substantial loss of dentin; no exposure of pulp	More than 2.0 mm deep defect (pulp may be exposed)
5.	Pulp exposure	Pulp exposure	Pulp exposure

Table 26.2: Simplified scoring criteria for tooth wear index

Score	*Criteria*
0	No wear into dentin
1	Dentin just visible (including cupping) or dentin exposed for less than one-third of surface
2	Dentin exposure greater than one-third of surface
3	Exposure of pulp or secondary dentin



Fig. 26.1: Severe attrition involving dentin



Fig. 26.2: Pathological attrition



Fig. 26.3: Attrition of occlusal surfaces with protrusion of amalgam

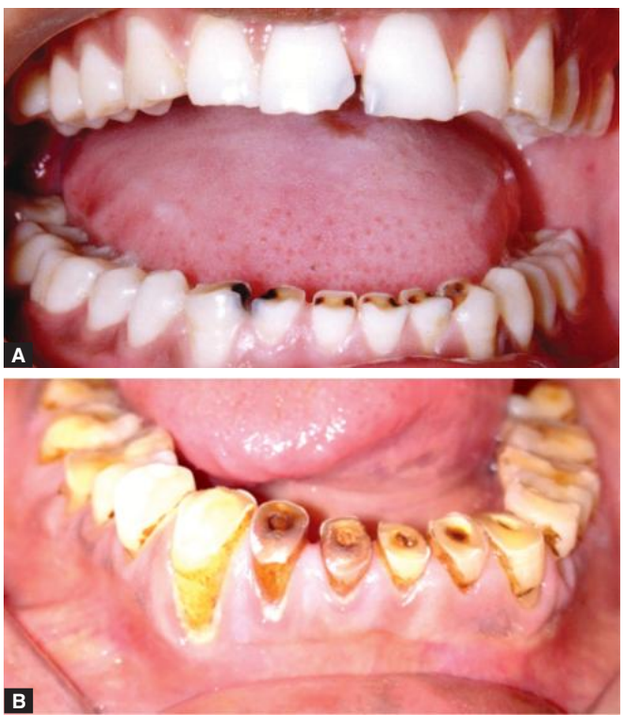


Fig. 26.4: Caries superimposing the attrited incisors

Modified index for assessing attrition

0 = No wear
1 = Minimal wear
2 = Noticeable flattening parallel to the occluding planes
3 = Flattening of cusps or grooves
4 = Total loss of contour and/or dentin exposure when identifiable

Caries can also superimpose at the attritional areas because of increased vulnerability of exposed dentinal surfaces (Fig. 26.4A, B).

It is established that the occurrence and pattern of tooth wear is related to educational, cultural, dietary, occupational and geographic factors. Other important factors are age and the function of occlusion.

The intensily of attrition has been used in archaeology and the forensic sciences to estimate human age. The canine-guided occlusion has significant influence in



Fig. 26.5: Deep bite



Fig. 26.6: Edge-to-edge bite

addition to crowding, occlusal slides, crossbites, chewing habits and diet (Figs 26.5 and 26.6).

It is hypothesized that enamel wear is affected by changes in lubricating conditions (acidity) in the oral cavity and the functional loads. It is also noted that attrition is a continuing process. It has been established that teeth continue to erupt in adulthood even in the absence of masticatory function and concomitant attrition.

BIOCORROSION

The term 'biocorrosion' embraces the chemical, biochemical, and electrochemical degradation of tooth substance caused by endogenous/exogenous acids, proteolytic agents as well as the piezoelectric effects. The definition of erosion does not account for proteolysis and piezoelectric effects, which may also be involved in the degradation of tooth substance; better defined by the term 'biocorrosion'.

Biocorrosion is regarded as one of the mechanisms for development of non-carious cervical lesions.

The term 'erosion' accounts for degradation of tooth tissues only because of action of acids. Erosion does not recognize the proteolysis and piezoelectric effects, which are also involved in the degradation of tooth tissues. The term 'biocorrosion' was conceived, which implies chemical/biochemical/electro-mechanical degradation of tooth tissues [chemical exogenous agents; biochemical proteolytic enzymes and electrochemical piezoelectric effect action on organic (collagen) matrix of dentin].

Erosion is considered as a physical mechanism causing wear because of friction from movement of liquids.

Since enamel is mostly inorganic, can easily be disintegrated by acids; whereas, dentin being mostly organic is degraded by proteolytic enzymes (protease). The enzyme is secreted by plaque micro-organisms through gingival cervicular fluid. The proteolytic enzymes may enter the oral cavity during diseases, like, gastroesophageal reflux disease, regurgitation and bulimia necrosa.

The collective mechanism of chemical action and the biochemical action may be termed as 'biocorrosion'. (Caries is considered as microbial biocorrosion; whereas, biocorrosion is affected by microbial invasion.)

The combined effect of biocorrosion and stress fatigue has also been reported. It is established that chemical and biochemical activity is accelerated during cyclic stress.

BRUXISM

Bruxism is a parafunctional activity of the masticatory system, which includes clenching and grinding at a subconscious level. Bruxism may occur during sleep or during wakefulness, commonly referred to as 'sleep bruxism' and 'awake bruxism', respectively.

Bruxism is classified as 'acute', 'subacute' and 'chronic', depending upon the patient's signs and symptoms. Bruxism is also classified as 'diurnal' and 'nocturnal', depending upon the activity timings. It is also referred as 'primary', where no systemic cause is evident; and 'secondary', where neurological/psychological factors are associated with bruxism.

Bruxism is a multifactorial condition. The most important etiological factor is the psycho-emotional status of the patient. However, bruxism is also related to the local, systemic and neurological features. Local factors include, trauma from occlusion, premature contacts, atypical eruptions, etc. Systemic factors include nutritional deficiency, gastrointestinal disorders, allergic reaction, mental retardation, etc.

Neurological factors include patients with neurological orders (neuroleptic), etc. Use of anti-depressants can also be the cause of sleep bruxism.

The management of bruxism includes occlusal adjustments and physiotherapy, such as, splints reduce symptoms, and distribute the masticatory forces.

CERVICAL LESIONS

Cervical lesions are the loss of hard tooth tissue at or around the cementoenamel junction of the tooth

crown/root. These can affect any surface of the tooth, i.e. facial, lingual or proximal. Defects arising on the cervical one-third of the facial and lingual surfaces of any tooth are included in Class V lesions. The term 'cervical lesion' should not be used to represent Class V lesions only (proximal cervical lesions will be Class II and III, as per the tooth). It is suggested that appropriate terminology be used whenever any reference is made to these lesions. Cervical lesions can be broadly classified into two categories:

I. Carious cervical lesions
II. Non-carious cervical lesions
   a. Abrasion lesions
   b. Erosion lesions
   c. Abfraction lesions

I. Carious Cervical Lesions

Caries is an important factor in the loss of tooth structure. Cervical carious lesions are smooth surface lesions, with their morphology similar to carious smooth surface lesions in any other region of the tooth (Fig. 26.7). These begin over a broad surface area and converge towards the dentinoenamel junction. On reaching the junction, the lesion again spreads laterally. Depending upon the stage of progression, lesions may present with variable morphology. Incipient defects are seen as chalky white opacities, which become visible only when the enamel is desiccated, and partially or totally disappear when the surface is hydrated (wet). These may lose their translucency because of the extensive subsurface porosity caused by demineralization. Their surface texture is unaltered and routinely cannot be detected by tactile examination with an explorer. At this stage, the lesion can be remineralized, if adequate preventive measures are taken. An advanced lesion is seen as a rough surface that is softer than the unaffected normal enamel. Further destruction of the tooth by caries results in frank cavitated lesions. The incidence and severity of attack increases several fold in conditions of rampant caries or radiation-induced xerostomia.



Fig. 26.7: Carious cervical lesions

II. Non-Carious Cervical Lesions

Many non-carious destructive processes include attrition, abrasion, erosion, abfraction, demastication and resorption. Amongst these, abrasion, erosion and abfraction are the common cervical lesions. It is emphasized to attain a detailed knowledge of the etiology of these defects for preventing further lesions and halting the progress of lesions already present. Also, treatment of these lesions will be ineffective unless the etiologic factor(s) are eliminated.

Non-carious cervical lesions, depending upon their etiologic factors, present a variable morphology, ranging from shallow grooves to broad dished out lesions to large notched/wedge-shaped defects, mostly on facial surfaces. Floor of the lesion may be flat, rounded or sharp angled. Rounded lesions are less frequently encountered than angular ones. The prevalence of non-carious cervical lesions, regardless of form and etiology, varies from 5 to 85%. Both prevalence and severity are known to increase with age. The older the patient population, larger is the percentage of individuals showing lesions, greater are the number of lesions in any one individual and deeper is the extent of lesions. The impact of these lesions on individual patients may also vary. Some patients present with no symptoms, while others may complain of highly sensitive teeth. Severe lesions may affect the vitality of pulp and threaten the structural integrity of the tooth.

Why Non-carious Cervical Lesions are usually on Facial Surface?

The relative thickness of bone on the lingual or palatal surfaces of teeth is consistently thicker than the facial surface. The bone may deflect the load differently on the facial surfaces than on the lingual surfaces of teeth. This can be termed 'osteodeflection' (from 'osteo' meaning bone and 'deflection' being the displacement of a structural element under load). When a lateral force vector from a facial to a lingual direction begins to tip the tooth lingually, the tooth cannot move bodily due to the thickness of the bone, but rather must bend at the fulcrum (Fig. 26.8A). In doing so, this places the lingual cementoenamel junction (CEJ) area under compression and the facial CEJ area under tension. This tensile load is far more damaging than compressive load. Conversely, if the force vector comes from a lingual to a facial direction, the tooth can tip bodily because of lesser bone thickness on the facial, thus not concentrating tensile stress at the lingual CEJ (Fig. 26.8B). Recession is more common on the facial and is generally accompanied by loss of vertical bone as well. This loss of bone on the facial allows the tooth to tip bodily in a facial direction, further reducing the

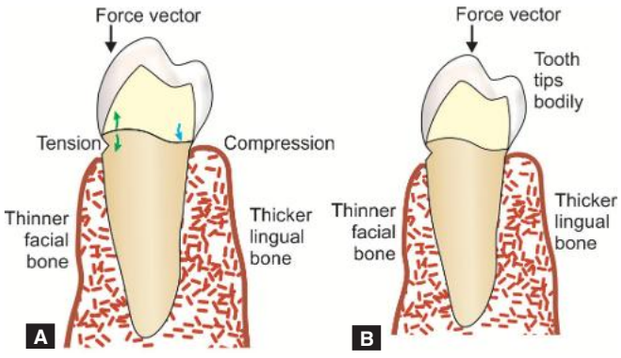


Fig. 26.8: **(A)** Direction of force from facial to lingual; **(B)** Direction of force from lingual to facial

tensile stress at the lingual CEJ. Conversely, forces that tend to move the tooth lingually are resisted by the lingual bone and the stresses at the facial CEJ can shift location. This change in the location of the fulcrum may cause the cervical notches to migrate incisally/occlusally or apically depending on the new site of the fulcrum.

The commonly encountered cervical lesions are as follows:

A. Abrasion

Abrasion is defined as "*wearing away of the tooth substance because of grinding, rubbing or scraping caused by external mechanical means, like in repeated contact of the teeth with foreign objects or substances*". Abrasion may affect one or more teeth, or the entire dentition. Any surface of the tooth may be affected but the most frequently affected site is the cervical region.

The cervical abrasion lesions can be due to:

- Faulty oral hygiene practice, like horizontal brushing, excessive forces applied during brushing, coupled with excessive time and frequency.
- Quality of tooth brush; hardness and shape of tooth brush bristles, flexibility and length of the tooth brush handle affecting the grip and also pH and amount of dentifrice used (Tooth powder is generally five times more abrasive).
- Ill-fitting clasps of partial dentures are also known to induce localized abrasion lesions.
- Cervical abrasive wear on the proximal surfaces is often caused by friction from objects, such as, toothpicks, interproximal brushes, etc.

Generally, enamel is hard and not easily abraded; therefore, it serves as a protection for the underlying dentin, which is abraded 25 times faster.

Cementum is the softest of all tissues; shows abrasion 35 times higher than enamel. This explains for the higher prevalence and severity of cervical lesions in the older age. With increasing age, gingival recession exposes the cementum and as a result the tooth becomes highly susceptible to abrasion even under the previously non-damaging oral hygiene measures.

Cervical lesions, because of abrasive forces, generally have sharp defined margins and hard smooth surfaces with burnished appearance. Hypersensitivity is intermittent in character, appearing or disappearing at intervals. In slow progressive defects, reparative dentin formation occurs over a period of time, making them asymptomatic. Lesions may show varying grades of depths like: (i) Shallow (S)—0.1–0.5 mm in depth, (ii) Deep (D)—more than 0.5 mm but no pulp exposure, and (iii) Exposure (E)—pulp is exposed.

Abrasion lesions usually affects facial surfaces of maxillary teeth; whereas, lingual surfaces are rarely affected. The canines and the premolars exhibit the highest frequency.

A survey was conducted on 460 patients who visited Government Dental College and Hospital, Amritsar, (India). 11,040 teeth were examined for cervical abrasion lesions. 2760 teeth were found to be cervically abraded and the percentage of lesions increased with age: 9.34% (31–40 years), 30.43% (41–50 years), 42.02% (51–60 years), and 18.18% (61–70 years). Males were comparatively more affected than females. The frequency with which the lesions occurred was in the following decreasing order of the affected teeth—upper and lower first premolars, upper canines, upper and lower second premolars, lower canines and incisors, and molars. Premolars were more susceptible to abrasion, being placed slightly protruded in the dental arch.

Features affecting process of abrasion are:

a. ***Effect of tooth brushing techniques:*** Brushing technique has been considered to be an important factor in developing abrasion lesion. The horizontal brushing technique significantly increases the frequency of cervical lesions as compared to conventional vertical/roll technique. Horizontal brushing usually produces V-shaped notches in dentin, independent of bristle stiffness. Cervical lesions are also associated with increased brushing frequency. Patients who brush quite often show more lesions irrespective of age group. The length of the time spent and force applied during brushing may also influence the development of lesions, e.g. right- handed individuals experience lesions on the left side and vice versa. This may be because of easier access and the force being applied on opposite side of the dental arch. The sequence in which teeth are brushed influences the location of lesions: More lesions are present in the quadrants brushed first. This might be because of the greater force being applied in the earlier tooth brushing

Table 26.3: Eccles index for Erosion Lesions

Class	*Surface*	*Criteria*
Class I		Early stages of erosion, absence of development ridges, smooth, glazed surface occurring mainly on labial surfaces of maxillary incisors and canines
Class II	Facial	Dentin involved for less than one-third surface: Two types; Type 1 (commonest): Ovoid-crescentic in outline, concave in cross-section at cervical region of surface. Must differentiate from wedge-shaped abrasion lesions. Type 2: Irregular lesion entirely within crown. Punched out appearance, where enamel is absent from floor
Class IIIa	Facial	More extensive destruction of dentin, affecting anterior teeth particularly. Majority of lesions affect a large part of the surface, but some are localized and hollowed out
Class IIIb	Lingual or palatal	Dentin eroded for more than one-third of the surface area. Gingival and proximal enamel margins have white, etched appearance. Incisal edges translucent due to loss of dentin. Dentin is smooth and anteriorly is flat or hollowed out, often extending into secondary dentin
Class IIIc	Incisal or occlusal	Surfaces involved into dentin, appearing flattened or with cupping. Incisal edges appear translucent due to undermined enamel; restorations are raised above surrounding tooth surface
Class IIId	All	Severely affected teeth, where both labial and lingual surfaces are extensively involved. Proximal surfaces may be affected; teeth are shortened

episodes. The amount of force used during brushing is determined by a number of factors, like the age of the individual, brushing technique, hardness of bristles and individual habits.

b. ***Effect of dentifrices:*** The adverse effects of abrasive toothpastes and powders have been established. The abrasive potential of a material varies. The amount of dentifrice applied, the degree of salivary dilution, technique and force applied during tooth brushing, all influence abrasion; thus not just a single factor, but multiple factors work together to produce a lesion.

Classification

Abrasion lesions, on the basis of morphology and location, may be classified as:

i. *Notch (N)/V-shaped defects:* Where oblique occlusal and cervical walls intersect at a certain depth with no definite axial wall in between them.
ii. *C-shaped defects (C):* Where cross-section of the defect is C-shaped with rounded floors.
iii. *Undercut concave (UC):* Where occlusal and cervical walls intersect with a definite axial wall in between them.
iv. *Divergent box (DB):* Where a definite axial wall is present with the occlusal and cervical walls diverging toward the surface.

B. Erosion

Erosion is defined as *"the defects arising because of dissolution of tooth structure subsequent to chemical attack of either endogenous or exogenous origin, or combined chemico-mechanical attack."*

Eccles (1979) classified erosion lesions (Table 26.3) into three classes.

Bartlett, et al (2008) introduced Basic Erosive Wear Examination (BEWE) to facilitate quantifying the risk of erosion. It is used as criteria of current indices and allows re-analysis of results from existing studies. All teeth, except third molars, are examined from the buccal, occlusal, and palatal aspects for acid damage. The most severely affected surface in a sextant is recorded with a 4-level score (Table 26.4). The sum of the scores defines the severity of erosive wear and guides management of the condition (Table 26.5). The maximum score per subject is 18. Management includes identification and elimination of the main causative factor(s), prevention and monitoring, as well as symptomatic and operative intervention.

Vialati and Belser (2010) introduced clinical erosive classification emphasizing on the maxillary

Table: 26.4: Basic Erosive Wear Examination: Criteria for grading erosive wear

Score	*Criteria*
0	No erosive tooth wear
1	Initial loss of surface texture
2[a]	Distinct defect, hard tissue loss <50% of the surface area
3[a]	Hard tissue loss ≥50% of the surface area

[a] with scores 2 and 3, dentin is often involved.

Table: 26.5: Clinical management based on BEWE scores (complexity level)

Susceptibility level	*Cumulative score of all sextants*	*Management*
Minimal	Less than or equal to 2	Routine maintenance Periodic evaluation: 2–3 years
Low	3–8	Oral hygiene and dietary advice. Identify the causative factor(s) and develop strategies to eliminate the cause. Periodic evaluation: 1–2 years
Medium	9–13	As above, and Measures to increase the resistance of tooth surfaces. Monitor erosive wear; placement of restoration can be delayed Periodic evaluation: 6–12 months
High	14 and more than 14	As above and In cases of severe progression, restorations can be planned Periodic evaluation: 3–6 months

Table 26.6: Vialati and Belser classification and management of erosion

Class	*Palatal enamel*	*Palatal dentin*	*Incisal edge length*	*Facial enamel*	*Pulp vitality*	*Suggested therapy*
Class I	Reduced	Not exposed	Preserved	Preserved	Preserved	No restorative treatment – prevention only
Class II	Lost in contact areas	Minimally exposed	Preserved	Preserved	Preserved	Composite restoration
Class III	Lost	Distinctly exposed	Lost (less than 2.0 mm)	Preserved	Preserved	Length enhancement
Class IV	Lost	Extensively exposed	Lost (greater than 2.0 mm)	Preserved	Preserved	Sandwich restoration
Class V	Lost	Extensively exposed	Lost (greater than 2.0 mm)	Distinctly reduced/lost	Preserved	Sandwich restoration
Class VI	Lost	Extensively exposed	Lost (greater than 2.0 mm)	Lost	Lost	Sandwich restoration

anterior teeth. This classification system not only assesses the severity of hard tissue loss but also provides a guide to the clinician on how to appropriately restore the affected teeth. The classification (Table 26.6) establishes six levels of wear according to the level of dentin exposure in the palatal areas, the preservation of the incisal edges, the length of the remaining clinical crown, the preservation of enamel on the labial surfaces, and the vitality of the pulp. The 'sandwich approach' as listed in Table 26.6 refers to the application of a resin-based material to treat the palatal surface wear, followed by the application of a labial/facial ceramic veneer.

Etiology

A. Depending upon the source of chemicals (usually acids), erosion may be intrinsic or extrinsic:
   a. *Intrinsic (endogenous) erosion*
      i. Gastric reflux—esophagitis, hiatus hernia, increased gastric pressure, neuromuscular diseases, vomiting—psychosomatic, anorexia, bulimia, drug induced and regurgitation
      ii. Faulty metabolism
      iii. Local acidosis
   b. *Extrinsic (exogenous) erosion*
      i. Environmental/occupation—wine tasters, swimmers
      ii. Dietary—soft drinks, acidic food
      iii. Medication—vitamin C, mouth washes
   c. *Idiopathic (unknown)*

B. Erosion is also categorized depending upon clinical severity and pathogenic activity of progression:
   a. Clinical severity on tooth surface
      i. Superficial lesion
      ii. Localized lesion

b. Pathogenic activity of progression
   i. Manifest or active
   ii. Latent or inactive

a. ***Intrinsic erosion:*** It is a result of endogenous acids and may be seen in situations like:

i. Gastric acid contacting the teeth during recurrent vomiting, regurgitation or reflux. Such a condition is evident in psychosomatic disorders, e.g. nervous vomiting, self-induced anorexia nervosa, bulimia, etc; or in pregnancy, somatic disorders like, gastric dysfunction, chronic constipation, duodenal and peptic ulcers, etc. This type of intrinsic erosion is most likely seen on palatal surfaces of anterior teeth, especially the maxillary teeth (Fig. 26.9). Individuals having these disorders might be consuming diets rich in citrus fruits, juices and acidic beverages, which may be an additional contributing factor.

Intrinsic factors associated with recurrent vomiting/persistent gastroesophageal reflux (Table 26.7) *are:*

- Alimentary tract disorders (peptic ulcer, chronic gastritis, intestinal obstruction)
- Neurologic disorders (migraine, diabetic polyneuropathy)
- Metabolic and endocrine disorders (adrenal insufficiency)
- Drug side effects (central emetic effects, gastric irritation)
- Psychosomatic disorders (stress-induced vomiting, eating disorders)

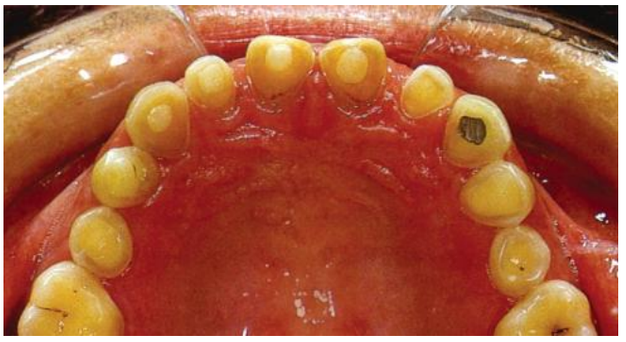


Fig. 26.9: Erosion involving palatal surfaces

Table 26.7: Gastroesophageal reflux disease

Symptoms	
Adults	*Children*
Acid taste in mouth	Feeding problem
Vomiting	Laryngitis/bronchitis
Sore throat	Asthma
Excessive salivation	Recurrent pneumonia
Halitosis	Gastric pain

ii. *Faulty metabolism:* Diseases, because of lack of oxygen or faulty metabolism, resulting in excessive excretion of acid sodium phosphate and/or acid calcium phosphate from the labial/buccal mucous glands. This type of erosion usually is confined to the labial surfaces of anterior teeth (Fig. 26.10) though other surfaces of entire dentition may be involved (Figs 26.11 and 26.12).



Fig. 26.10: Erosion involving labial surface of maxillary right central incisor



Fig. 26.11: Erosion involving mandibular premolars



Fig. 26.12: Multisurface erosion

iii. *Local acidosis:* Acidity from a local acidosis in periodontal tissues as a result of traumatic occlusion.
Excess of acidic salts present in the saliva from fermentation process.

b. ***Extrinsic (exogenous) erosion:*** It is a consequence of exogenous acids found as a contaminant in the work atmosphere (e.g. acid battery factories) or in chlorinated swimming pools, in low pH medications, like iron tonics, aspirin, hydrochloric acid replacements, oral hygiene products and in the diet. The most commonly consumed, potentially damaging acidic diets are, phosphoric acids in fruit juices, lemons, vinegar and beverages (Tables 26.8 and 26.9). Ascorbic acid/citric acid added to a variety of drinks and candies can also be a possible cause for erosion. The citrate ions are believed to bind with calcium in enamel and dentin forming soluble calcium citrate. The diets, which include the consumption of more fruits and vegetables are an important factor in the etiology of dental erosion. Also, during fasting, the combination of acidic drinks and reduced salivary flow contribute to increased risk of erosion. Such lesions frequently present on the facial surfaces of anterior teeth. Fruits seem to affect anterior teeth while fruit juices may affect premolars and molars. Cervical surfaces are most prone areas as they are close to the gingiva and less cleansable; foods and beverages may harbour in their proximity for longer periods of time. Erosion, because of air-borne acids, is comparatively rare at the cervix. It is important to differentiate erosion lesions from carious lesions, the latter are characterized as a disease that occurs by the action of acids produced by plaque micro-organisms; whereas, the former result from direct contact with acids of any origin except micro-organisms.

Table 26.8: Acidity of common foods and beverages

Pepsi/coke	2.5–2.7
Fruit beer	3.0
Orange juice	2.8–3.8
Lemon juice	1.8–2.4
Pickles	2.5–3.0
Beer	4.0–5.0
Wine	2.2–3.5
Sauces	3.5–4.5

Table 26.9: Factors influencing erosive potential of foods and beverages

Chemical	pH and buffering capacity Type of acid, adhesion to teeth, concentrations of acid
Behavioral	Drinking habits, excessive consumption of acid foods, oral hygiene measures
Biological	Salivary flow, buffering capacity, acquired pellicle, soft tissue movements

c. ***Idiopathic (unknown source of erosion):*** It has also been reported; might be because of genetic disturbance or metabolic imbalance of the body fluids.

Erosion rarely operates alone in causing tooth loss. Erosion lesion is multifactorial, i.e. once the surface of the tooth is hypomineralized by erosion, wear resistance of dental hard tissues is lessened and they become more prone to damage by mechanical abrasion, demastication (abrasiveness of individual foods), friction from labial and buccal muscles and possibly abfraction. It is critical to differentially diagnose the lesion before planning treatment.

Adjunct Etiological Features

The development of dental erosion is strongly influenced by:

A. Determinant factors
   a. Biological factors
   b. Behavioral factors
   c. Chemical factors

B. Modifying factors
   a. Knowledge
   b. Systemic health
   c. Socioeconomic status

A. Determinant Factors

a. Biological factors

i. ***Saliva:*** Saliva plays an important role in modifying the erosive effects of dietary foods and beverages; the possible mechanisms are:
- Dilution and clearance of an erosive agent from the oral cavity
- Neutralization and buffering of dietary acids
- Formation of a pellicle layer on the surface of enamel, which protects it from demineralization by dietary acids, while enhancing remineralization by providing calcium, phosphate and fluoride to eroded enamel and dentin.

Both quantity and quality of saliva control the extent of dental erosion. Individuals with decreased salivary flow and buffering capacity, erosion is expected to be higher. It has been suggested that saliva rich in mucin prevents precipitation of calcium phosphate that is beneficial in managing acid injuries to enamel.

ii. ***Acquired pellicle:*** Acquired pellicle is a protein-based semipermeable layer, rapidly formed on the tooth surface, shortly after tooth brushing, prophylaxis, or chemical dissolution. This pellicle acts as a barrier, which prevents the direct contact of erosive agents on the tooth surface and serves as reservoir of remineralizing electrolytes. One should not brush teeth immediately after consuming acidic drink, because it removes the acquired pellicle, thus leaving the teeth less protected.

iii. ***Tooth structure:*** The developmental pattern of erosion is influenced by the clinical history of each patient. The situations that can impair or modify the tooth, include caries, erosion, abrasion, attrition, and abfraction. With regards to tooth composition, exposures to saliva and fluorides may help remineralization of eroded enamel (Formation of fluoroapatite may influence the development of dental erosion).

iv. ***Soft tissues:*** Oral soft tissues and their movement influence the tooth sites that will contact acidic substances and affect clearance of acids in their proximity. Soft tissues may also promote tooth wear. Erosion on palatal surfaces of teeth frequently touched by tongue has been reported.

b. ***Behavioral factors:*** Behavioral patterns influence the biological response to erosive challenges, modifying the oral environment by making it more or less susceptible to development of erosion. Behavioral aspects influence the intensity, localization and type of erosion lesions.

i. ***Unhealthy lifestyles:*** Individuals consuming alcohol may be at risk of developing erosion due to symptoms, such as recurrent vomiting and gastrointestinal reflux. The use of drug, such as 'ecstasy' (3,4-methylenedioxy-methamphetamine) is known for causing tooth wear. Ecstasy users report symptoms, such as dry mouth, hyperthermia, clenching, and grinding; indicating that the physical wear may continue for some time after the drug consumption.

ii. ***Healthy lifestyles:*** It is emphasized that people involved in sports and exercise may be at risk of developing erosion due to the consumption of sport drinks, replenishers, fruit juices, and other acidic beverages. Exercise increases the loss of body fluids and may lead to dehydration and a reduction of decreased salivary flow can create an environment for the development of erosion.

iii. ***Nutritional habits:*** The frequency of acidic consumption along with prolonged contact of an acid with tooth surface increases its damaging potential. Certain bizarre habits can modify the risk effects, e.g. a beverage may be swished or rinsed inside the oral cavity before final swallowing. Habits, such as lemon sucking and soft drink swishing, expose the tooth surface for a longer period of time, which may cause greater demineralization. Phosphoric acid, usually found in soft drinks, is three times more erosive than any organic acid (citric, tartaric, maleic, lactic acids etc.). Citric and maleic acids are predominantly found in fruits and in their derivate products. Tartaric acid is present in grapes and wines.

Roughness of foods also increases wear of the affected tooth surface. This can be particularly evidenced in lactovegetarian diets (healthy and rough foods), which show a greater incidence of erosive lesions.

The diet analysis and recommendations for individuals with erosion is summarized in Table 26.10.

c. ***Chemical factors:*** Chemical factors imply parameters inherent to erosive beverages, food, and other products. In erosion, the amount of mineral dissolved from enamel depends upon the pH, buffering effect and the amount of calcium, phosphate, fluoride, etc. present in the drink. As

Table: 26.10: Diet (nutrition) analysis and recommendation for individuals with erosion

	Diet (nutrition analysis)	*Recommendation*
Water	Do you drink enough water?	Drink 2.5 litres water per day (enough water especially for labourers and sportsperson)
Acidic drinks	Do you drink excess soft drinks?	Avoid acid drinks in routine
Fresh fruits (diet)	Do you eat fresh fruits daily?	Fruits stimulate saliva; take plenty of fruits
Calcium content in diet	Are you consuming enough calcium in your diet?	Milk, cheese and yoghurt contain sufficient calcium; protect teeth against acids
Lifestyle	Do you have a healthy lifestyle?	Excess alcohol is dehydrating; may cause gastric acid reflux
Health	Do you have any health problem?	Drugs, given for asthma, depression, hypertension, etc. reduce salivary flow

the pH decreases, solubility of enamel apatite increases. At a pH 3.0, solubility is 75 g/L; rises sharply to 400 g/L at a pH 2.5. When the pH is around 4.0, it is possible to counteract enamel dissolution by addition of calcium phosphate to the drink. Below this pH, the solubility of enamel apatite increases steeply; calcium phosphate supplementation in counteracting enamel dissolution is ineffective. The ability of an acidic solution to dissolve enamel apatite depends upon its capability of maintaining the pH level from being affected by dissolution of apatite and dilution with saliva. This property is known as the *'buffering effect'*.

B. *Modifying Factors*

a. ***Knowledge:*** A thorough knowledge of risks and activity of erosion, as well as an understanding of the erosive potential of drinks and foodstuff can minimize the initiation and progression of such lesions. Patients usually are not aware of the erosive characteristics of food items and the associated modifying factors.

b. ***Systemic health:*** Certain systemic diseases (Sjogren's syndrome), medicines (acetylsalicylic acid, diuretics and anti-depressive medicines), and therapies that involve irradiation of salivary glands; all adversely affect salivary production, interfering with the biological protection provided by saliva. Hyposalivation may induce dental erosion without excessive exposure to erosive agents.

c. ***Socioeconomic status:*** Socioeconomic status may be linked to systemic health and knowledge; thus indirectly influence development of dental erosion.

Clinical Implications

Erosion lesions are mostly broad, shallow, saucer-shaped depressions present in enamel and/or dentin; with no sharp line angles and less well-defined margins. At times, lesions may even be grooved, wedge shaped or irregular. When the dentin is exposed, tubular calcification occurs along with discoloration. Hypersensitivity may be marked in rapid forms or in lesions where dentin is exposed with no underlying tubular calcification. In some instances, caries may supervene and the characteristic features of erosion are lost.

The features of importance are:

i. ***Decrease in enamel microhardness:*** The episodic exposure to acidic challenges decreases enamel microhardness, which upon exposure to mechanical forces, makes the surface more susceptible to disruption.

ii. ***Formation of reactionary dentin:*** The dentin–pulp complex responds by producing reactionary dentin and occluding the dentinal tubules in order to compensate for the tissue loss. If the progression of dental erosion surpasses the dentin–pulp complex reparative capacity, it may lead to toothache, dental sensitivity and pulpal inflammation; subsequently pulp necrosis.

How Enamel and Dentin React to Attack by Erosive Agents

Dental erosion is believed to be a result of demineralization of the inorganic matrix of the tooth.

Eroded enamel appears smooth and polished; perikymata are usually absent and it may or may not be discolored. The hydrogen ions or chelating agents start to dissolve the enamel crystal. First, the prism sheath area and then the prism core are dissolved, leaving a honeycomb appearance. Then, fresh and unionized acid will diffuse into the interprismatic areas of enamel and further dissolve the mineral content underneath the surface.

Usually, two erosive phases are seen on enamel. One is *'manifest erosion'*, which appears as a hollowed out pitted surface resembling honeycomb. The pitted surface characterizes ends of enamel prisms that have been dissolved below the level of interprismatic matrix. Second is *'latent erosion'*, which is inactive stage and the prisms are less obvious. 'Manifest erosion' is more common than 'latent erosion' and is seen more frequently in females and young individuals.

The initial events in dentin demineralization are similar to those in enamel. However, because of the high organic content of dentin, diffusion of the demineralizing agents are partially stopped by the organic matrix, which acts as a barrier to acid diffusion. The erosive process in dentin is not affected by this phenomenon; the dentin substrate is more susceptible to acid dissolution because its hydroxyapatite crystals are smaller than that of enamel. Peritubular and intertubular dentin may also be demineralized laterally along the open tubules.

C. Abfraction

Abfraction is defined as *"wedge-shaped defect in the cervical region of the tooth and is hypothesized to be a result of tensile stresses concentrated in this area consequent to occlusal forces in some remote area"*. The term 'abfraction' was first coined by Grippo (1991) to distinguish the lesion from abrasion and erosion. A few authors have also termed these lesions as *'idiopathic cervical erosions'*.

It is hypothesized that the tensile stresses may have role in developing cervical abfraction lesions.

Normally, during mastication, when moving from working side to centric occlusion, lingual slopes of the maxillary cusps contact buccal slopes of the mandibular cusps. This contact serves as an inclined plane and forces are generated perpendicular to the tangents drawn from the respective cusps. When the eccentrically lateral forces are resolved into two components, the vertical component is directed along the long axis of the tooth (well tolerated, because it is compressive in nature); whereas, horizontal component is perpendicular to the long axis. The transverse force is responsible for creating deflection/flexure in the tooth structure, i.e. the tooth is compressed primarily on the side towards which it is being bent and is subjected to tensile stress on the side away from the direction of bending. For example, in lingually directed forces in a mandibular molar, lingual portion of the tooth is compressed, while the buccal portion is stretched, with the fulcrum at the cemento-enamel junction. The region under greatest tensile stress is close to the fulcrum, while regions of greatest compressive stress are the occlusal contacts, and the apex of the root.

Generally, these forces in ideal occlusion create deflection, which is within the tolerable limits of the tooth. However, the magnitude of transverse force (shown for buccal cusps in Fig. 26.13) and the consequent bending movement increases with excessive cuspal slope and/or lingual inclination of the mandibular teeth. Also, masticatory forces in individuals with hyper- or malocclusion and parafunctional forces in bruxism may expose one or more teeth to strong lateral forces beyond the capacity of the teeth to withstand, resulting in cervical lesions (Fig. 26.14). For example, in dentitions that lack canine disclusion, lateral forces are transmitted to the posterior teeth during excursive

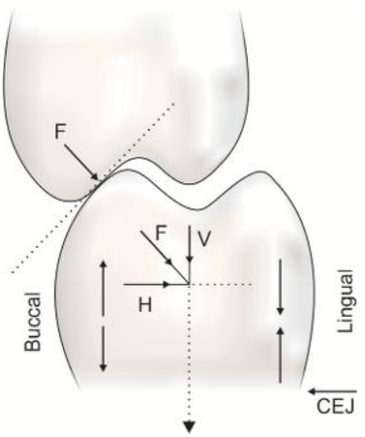


Fig. 26.13: Applied force on a cuspal incline (F) resolved into its vertical (V) and horizontal (H) components produce compression on the lingual side and tension on the buccal side



Fig. 26.14: Caries superimposed on abfraction lesions in bruxism

movements, thereby raising the chances of developing abfraction lesions. In bruxists, if canine disclusion is absent, the incidence is further increased many fold. Abfraction lesions are mostly seen on the buccal surfaces of mandibular teeth, as evident in Fig. 26.13 (Buccal surfaces of maxillary teeth will not stress to tension, rather placed under compression).

How Abfraction Lesions Develop

The masticatory system during function exposes the teeth to three types of stresses: compressive, tensile and shear. Both dentin and enamel have high compressive strengths; however, weak in tension (Enamel is 35 times stronger, and dentin is 7 times stronger in compression than in tension). Dentin is substantially stronger than enamel in counteracting tensile stresses. The high resiliency and elasticity enables it to withstand greater deformation without fracture. Enamel is made up of three components: A mineral component hydroxyapatite in enamel rods, an organic matrix and water (free or bound). The ability of enamel to withstand stresses depends significantly on the direction of forces in respect to orientation of enamel rods. Forces are best withstood when they are in line with the rods; whereas, under tensile forces that tend to pull the rods away from each other, enamel surrenders. There occurs disruption of bonds between hydroxyapatite crystals. As bond is broken, small intercrystalline spaces are created, which allow penetration of water and other molecules. The disrupted crystalline structure is susceptible to chemical dissolution and breakage from physical forces, like brushing, compression, tension, mastication and bruxism (Fig. 26.15).

Other possible mechanisms that contribute to loss of tooth tissue as a consequence of occlusal loading are: (a) When a tooth is subject to tension or compression, electric potential gradient is generated between different regions of the tooth. The so-called piezoelectric effect may promote loss of tooth substance. (b) The

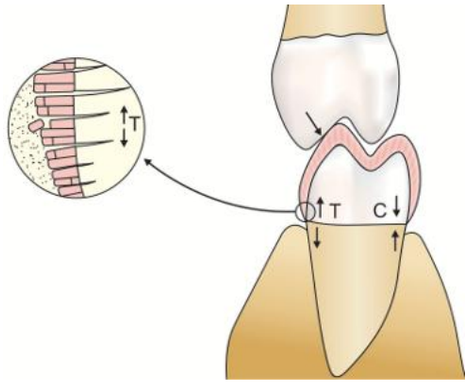


Fig. 26.15: Lateral forces create tension (T) and compression (C) at the cervix as indicated by arrows. In the enlarged section, disrupted bonds between enamel rods are shown

thickness of enamel and dentin varies over the cusps of posterior teeth, causing variation in the contour of dentinoenamel junction. A concavity on the dentinoenamel junction is usually found in the occlusal third of the facial cusps of mandibular premolar teeth and lingual cusps of maxillary premolar teeth. The location of this concavity corresponds to the most common functional location of premolar cervical lesions, i.e. below mandibular functional cusps and above maxillary nonfunctional cusps. When this concavity was eliminated in a finite element analysis, tensile stresses were also eliminated in the corresponding cervical area. This hypothesized that the cuspal contour of dentinoenamel junction had a role to play in determining the amount of stress produced at the cervix.

Characteristics of Abfraction Lesions

The characteristic features of abfraction lesions are:

- It is at or near the fulcrum.
- A wedge-shaped lesion with sharp line angles. Local factors, like abrasion, erosion, demastication, etc. may modify the shape of lesion, but the overall pattern remains wedge shaped.
- The direction of lateral forces determines the location of lesions, i.e. the number of lesions that can develop on the same tooth depends upon the number of directions of lateral forces, e.g. if there are two differently directed lateral forces acting on the same tooth, the final lesion would be a combination of two lesions generated by each of the two forces (Fig. 26.16A, B).
- The size of the lesion is determined by the magnitude and frequency of applied tensile force. The location of fulcrum determines the exact site of the lesion. For a given lateral force, magnitude of tensile stress is determined by the distance between the applied force and fulcrum. Greater this distance, larger is the tensile stress generated near the fulcrum and, therefore, larger the region of disruption in this region (Fig. 26.17). It is well known that opposing teeth rarely contact at a single point, instead they contact over a surface area. As a consequence, the force is distributed along the contact plane, and the distance of any one point of force from the fulcrum is different from any other point in the plane. This exposes a large area of the tooth to tensile stresses and the lesion so formed is dictated by the area of contact.

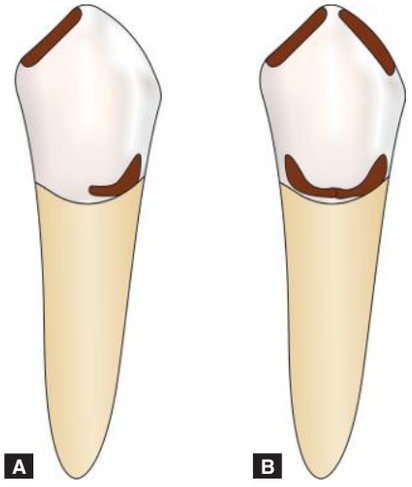


Fig. 26.16: Location of lesion as per direction of force: **(A)** Morphology of lesion as dictated by the contact plane; **(B)** Two differently directed lateral forces acting on the same tooth produce a combined lesion

A brief overview of characteristics of different lesion is presented in Table 26.11.

Arguments Relating to Tooth Flexure Theory

1. Greater prevalence of cervical lesions in older individuals may be because of increased brittleness of enamel and dentin with increasing age, which

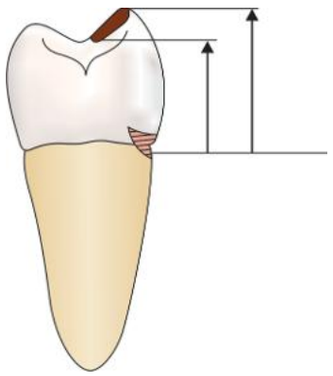


Fig. 26.17: Functioning along a contact plane: Magnitude of tensile stress on tooth is a function of distance between applied force and fulcrum

Table 26.11: Characteristics of different cervical lesions

Characteristic	*Erosion*	*Abrasion*	*Abfraction*
Location	Lingual or facial	Facial	Facial
Shape	U or dish shaped shallow	Wedged, notched or grooved	Single or overlapping wedge shape lesions
Margins	Smooth	Sharp	Sharp
Enamel surface	Smooth, may be polished	Smooth, may be scratched	Rough in initial stages, may have corrugated appearance

further exacerbates the effects of occlusal forces. Since damage to dental hard tissues is also a function of duration and magnitude of applied force, more lesions are likely to occur at older age.

2. More lesions are seen in bruxers compared to non-bruxers. Bruxism is a likely source of increased occlusal stress (80% of bruxers and 20% non-bruxers suffer from cervical lesions).
3. Increased loss of Class V restorations is seen in patients with bruxism, malocclusion or some form of stressful occlusion. Debonding occurs because of tensile stresses generated by lateral deformation of teeth.
4. Periodontal status may affect the development of abfraction lesions: Alveolar bone loss changes the fulcrum of bending moment causing the lesions to shift more apically. Conversely, loss of periodontal support leading to a high degree of tooth mobility may serve as a protective factor in that instead of flexing at the cementoenamel junction, the entire tooth moves away from the occlusal force.
5. Amalgam restorations on the occlusal surface are associated with increased cusp flexure and can weaken the tooth structure. Any factor that decreases the resistance of tooth to withstand stresses probably helps in lesion formation.
6. The theory of tooth flexure explains the development of lesions on isolated or non-adjacent teeth and in subgingival regions.

The dynamics of tooth substance loss is summarized in Flowchart 26.1.

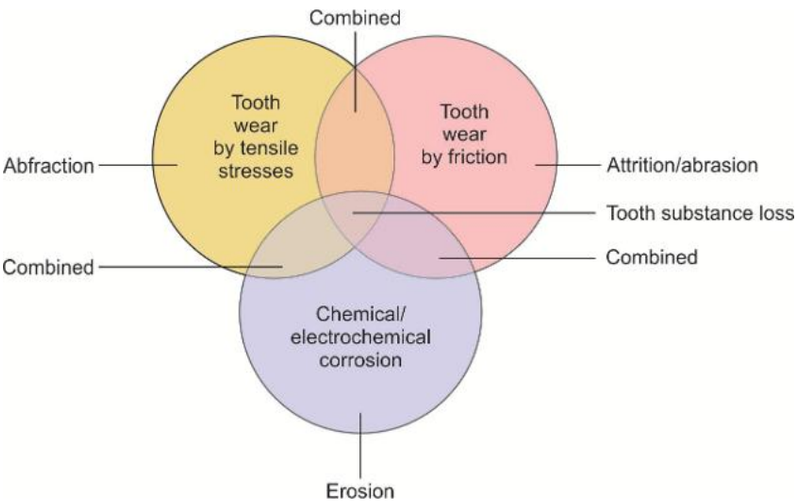
BIOMECHANICS OF A CLASS V CAVITY

The early authors were of the view that failure of cervical restorations might be due to pulling forces of the sticky food. Also, the extrusion of a Class V amalgam restoration was attributed to delayed expansion of amalgam due to water contamination. The hypothesis of delayed expansion could not explain all the features, because:

- Extrusion occurred even when rubber dam was used for placement of amalgam.
- Extrusion was commonly seen on the occlusal margin rather than on the wet gingival margin.
- Extruded restorations were confined to the buccal surfaces of mandibular premolars and molars.

These features hypothesize the role of biomechanical interactions in Class V lesions/cavity. Gabel was the first to explain the mechanical forces disrupting Class V restorations when the slopes of the two opposing teeth meet each other in lateral excursive movements, the contact serves as an inclined plane and forces are produced perpendicular to this plane. This force is resolved into two components: Vertical component parallels long axis of the tooth; while, horizontal component is perpendicular to the long axis. The tooth being firmly held in the alveolus, when

Flowchart 26.1: Dynamics of tooth substance loss



subjected to a transverse force, behaves like a cantilever beam. This places one side, i.e. buccal or lingual in tension and the opposite side in compression.

If a Class V cavity is prepared, the size of the preparation subjected to horizontal force (large size of preparation, more deflection) facilitates more deflection under a given load than when the tooth is unprepared. When subsequently a restoration is placed, it is also subjected to similar lateral deflecting forces. If the restoration is placed on the tooth side, which is under tension, interface at the occlusal margin tends to open up and a wedge-shaped defect is created between the tooth and the restoration (Fig. 26.18). Conversely, if it is placed on the side of the tooth being compressed, it would move out of the cavity because of pressure applied onto it (Fig. 26.19).

It has been established that there exists a minimum cavity size that does not alter the load deformation response of the tooth; however, this size has not been specified (large size, more deflection).

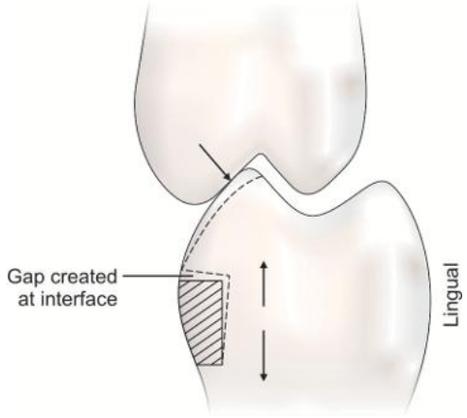


Fig. 26.18: Restoration being subjected to tensile forces; as a result, a gap is created at the interface

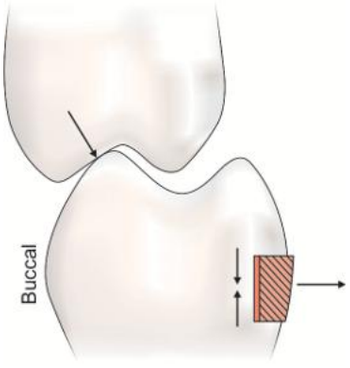


Fig. 26.19: Restoration being subjected to compressive forces and as a result, restoration may move out of the cavity

MEASURING TOOTH SUBSTANCE LOSS

The methods and the instruments measuring tooth wear are as follows:

a. Measurement *in vitro*

Since simulating oral environment is difficult, most of the experiments have been carried out *in vitro*. The *in vitro* examination has the advantage, such as variables, like exposure time and nature of the agent can be controlled.

The *in vitro* methods for measuring the tooth wear are:

i. ***Polarized light microscopy:*** Direct viewing under polarized light is helpful; trends of tooth wear can be ascertained.

ii. ***Microhardness:*** The technique assesses the hardness of the tooth before and after the erosion. This hardness data is qualitative and may not measure the exact extent of the tooth tissues loss.

iii. ***Surface profilometry:*** The loss of the tooth substance is measured by the surface profilometry by driving the spherical diamond tip onto the specimen. This method can measure actual depth of the lesion and also the dissolution of enamel.

iv. ***Microradiography:*** The subsurface demineralization of the area along with total mineral loss is being measured by microradiography, which is superior to surface profilometry.

v. ***Digital image analysis:*** The digital image analysis is useful in calculating the damage to the enamel. This technique is precise and sensitive and can be used both qualitatively and quantitatively. This technique measures the three dimensional loss of the tooth structure.

vi. ***Scanning electron microscope:*** The casts of the abraded or attrited teeth are examined under the scanning electron microscope. The impressions can also be examined by these methods. This method is precise, but only comparisons can be carried out at various intervals of time.

b. Measurement *in vivo*

The *in vivo* measurement of tooth substance loss is difficult, since the wear is different at different locations.

Certain indices may provide an idea of tooth wear; tooth wear indices are available to quantify tooth wear.

In vivo measurement can be indirectly carried out using study casts after silicone impressions. Study casts monitor the progress of tooth wear. Practically, tooth wear is measured using *in vitro* methods.

TREATMENT OF TOOTH SUBSTANCE LOSS

A. Treatment of Carious Cervical Lesions

Smooth surface lesions in the cervical portion of the teeth can be overlooked in a routine clinical examination, unless the teeth are isolated and dried and are free of debris. Radiographic analysis also provides little benefit. The depth of the lesion cannot be ascertained on the radiograph; also the radiolucency superimposes on the pulp, mimicking the appearance of a pulpal exposure. Management of cervical caries involves prevention and restoration. If caries is superficial, restrict to preventive measures; otherwise restoration.

a. *Preventive Protocol*

The preventive protocol includes:

i. *Diet modification:* The sucrose consumption should be limited and in-between snacks be avoided.
ii. *Shift oral microbial flora from pathogenic to non-pathogenic state* by using bactericidal mouth rinses like, chlorhexidine, topical fluoride applications, etc.
iii. *Periodontal treatment* is carried out to improve upon the gingival health. Oral hygiene should be maintained.
iv. *Modification of the tooth surface:* The resistance to demineralization is achieved by application of topical fluorides. Their continued use prevents development of new lesions and allows remineralization of initial lesions.
v. *Stimulating salivary flow* facilitates, clearing away of substrate/acids and promote buffering, etc. It is carried out by consuming food stuffs that require a lot of chewing, or chewing sugarless gum, etc.
vi. *Application of self-assembling peptide P11-4* (curodent repair) has shown significant improvement in the prevention of erosion from acid challenges. P11-4 can promote remineralization, effectively preventing erosive lesions.

b. *Restorative Aspects*

In case of rampant caries and radiation-induced xerostomia, etc. where multiple acute caries lesions are rapidly progressing, treatment needs to be initiated on an emergency basis. This includes removing infected tooth structure, and restoring the defects with temporary restorations. Most of the infecting organisms are removed with this technique and further lesion formation is prevented. The emergency regime must be accompanied by preventive measures, which further reduce and prevent micro-organism build up. Strict dietary counseling and intensive oral hygiene measures should be followed. The conventional caries lesions are restored with appropriate materials.

B. Treatment of Non-Carious Cervical Lesions

A detailed history is mandatory comprising of the following aspects: Dietary habits, gastric disturbances, acidic mouth taste, drugs, radiotherapy, salivary gland dysfunction, work-related exposure to acidic environments, parafunctional habits, oral hygiene measures, etc. Occlusion should be checked and a record of the study models made to monitor progression of lesions. After a thorough evaluation, management is carried out in two phases, i.e. preventive and restorative.

a. *Preventive Protocol*

The preventive protocol includes:

i. ***General preventive measures:*** These lesions may not be entirely prevented, but definitely reduced by regulation of diet, and parafunctional habits. Consumption of plain water reduces chances of erosion lesions.

ii. ***Local preventive measures***

- Use soft tooth brushes and low abrasive toothpastes, utilizing less force.
- Avoid ill-fitting metal clasps.
- Control parafunctional habits.
- Regulate frequency of consumption of acid foods and beverages.
- 'Drink only'; do not sip or swish acid beverages.
- Do not brush immediately following intake of acidic drinks.
- Use neutral pH fluoride mouth rinses daily.
- Relieve traumatic occlusion
- Enhance defense mechanism of the body by increasing salivary flow.
- Enhance acid resistance and remineralization potential by appropriate agents.

b. *Restorative Aspect*

The following factors should be considered prior to restoration:

i. *Structural integrity of the tooth:* If the notched/affected area is large/deep, strength of the root at the cervix is lessened. Bonded restoration is indicated in such cases.
ii. *Pulp protection:* When the lesion is deep, endangering the vitality of pulp, pulp capping procedures should be carried out.
iii. *Sensitivity:* When sensitivity continues to exist despite use of desensitizing agents, lesions should be filled.
iv. *Esthetics:* Esthetics should be considered before planning restoration of these lesions.

v. *Gingival health:* If the lesion is irritating the gingival tissue, the defect should be restored maintaining the gingival health.

vi. *Caries:* If caries supervenes the lesion, it should be restored; otherwise the incipient/superficial lesions can be treated by preventive measures alone.

vii. *Presence of removable partial denture:* If the location of the lesion interferes with the design of removable partial denture, restoration is indicated.

The non-carious cervical lesions are restored with materials having adhesion potential, viz. composites, compomers, glass-ionomers, resin-modified glass-ionomers and a combination of glass-ionomer and composite in sandwich technique. The material should be able to withstand degradation due to erosive attacks and toothbrush abrasion.

i. ***Restoration with composite:*** Restoration with composite resins offers the advantage of best esthetics and excellent bonding to etched enamel. Disadvantages include, exacting placement, and doubtful long-term bonding to dentin and cementum.

Since bonding to dentin or cementum is not as efficient as to enamel, microleakage and loss of retention is likely to occur at the gingival margin.

No instrumentation is desired in the non-carious cervical lesions when restoring with composite. It is demonstrated that V-shaped cavities are preferable over box-shaped cavities, as in the former the volume/ area ratio is less and hence the amount of polymerization shrinkage and subsequent gap formation is reduced. A small amount of instrumentation may be carried out on the incisal or occlusal margins with a tapering fissure bur to create a bevel on the enamel. This exposes enamel rod ends rather than lateral surfaces to the etching solution, thereby improving retention. Improving retention at the non-enamel (gingival) margins by placing slight undercut in dentin can be considered. A groove additionally resists the effects of polymerization shrinkage and tooth flexure, hence increasing resistance to microleakage. Polyacid-modified composites, i.e. compomers can also be used for restoration of non-carious cervical lesions. Their restoration process is similar to that for composite resins.

The final restoration is finished and polished (Fig. 26.20A, B).

ii. ***Restoration with glass-ionomer:*** Restoration with glass-ionomer offers the advantages of adhesion to enamel and dentin even in wet environments, and release of fluorides on a long-term basis. Glass-ionomers adhere to the tooth surface through hydrogen and ionic bonding; hence, mechanical retention is not required.

Since water is an important component in ionic exchange, the tooth should not be unduly dehydrated.

Placement of cement under moderate pressure is desirable to ensure optimum adaptation of the cement to the underlying tooth structure. Light-cure glass-ionomers are also used for restoration of non-carious cervical lesions. They overcome the disadvantages of conventional glass-ionomers by showing rapid initial set, and less sensitive to moisture contamination. The restoration is contoured and polished after one day and if possible, after one week (Figs 26.21A, B and 26.22A, B).



Fig. 26.20A: Abrasion lesion on maxillary canine and first premolar (preoperative)



Fig. 26.20B: Abrasion lesions restored (postoperative)

iii. ***Sandwich restoration:*** Restorations with combined glass-ionomer and composite is referred to as the 'sandwich or the laminate technique'. Glass-ionomer serves as a dentin replacement and provides chemical adhesion to the underlying tooth structure; whereas, composite provides control of contour and esthetics. The weak link in



Fig. 26.21A: Erosion lesion on mandibular canine and first premolar



Fig. 26.21B: Restored lesion

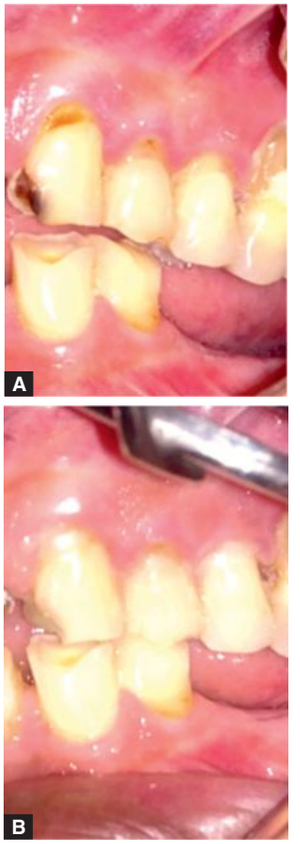
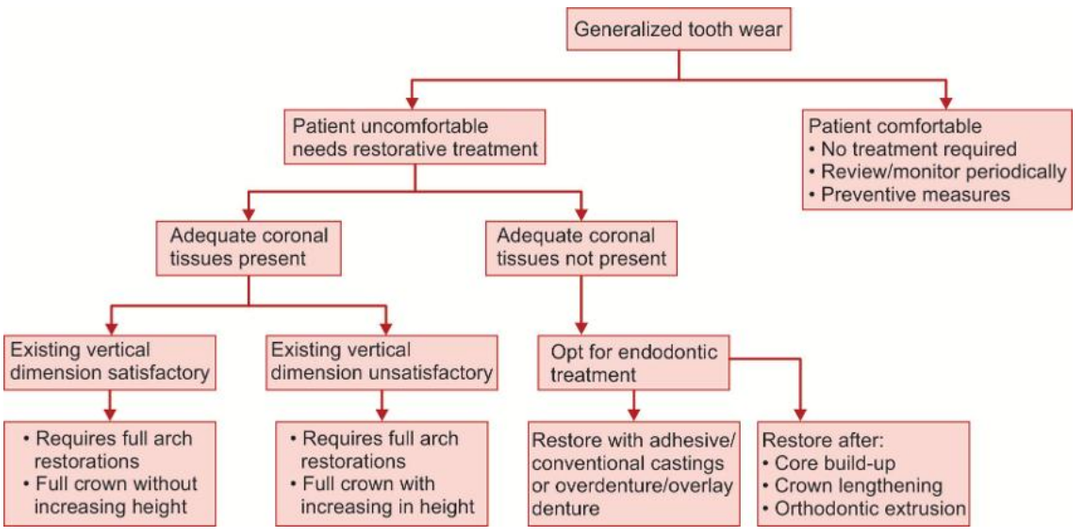


Fig. 26.22: **(A)** Abrasion lesions on maxillary, first and second premolars (preoperative), **(B)** Restored abrasion lesions, first and second premolars (postoperative)

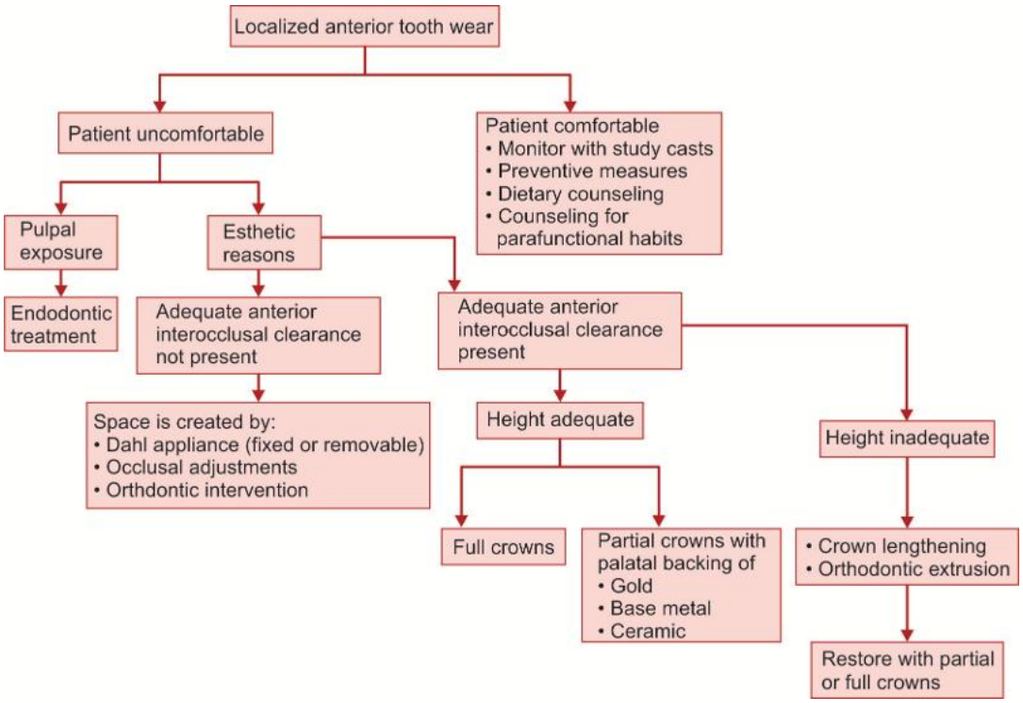
the dentin-glass-ionomer-composite bond is usually the glass-ionomer cement itself, i.e. failure is generally cohesive in nature. This necessitates use of glass-ionomer cement in adequate thickness.

In sandwich restorations, resin-modified glass-ionomers can be considered as these provide chemical attachment to the overlying composite resin compared to conventional glass-ionomers, which provide only mechanical attachment.

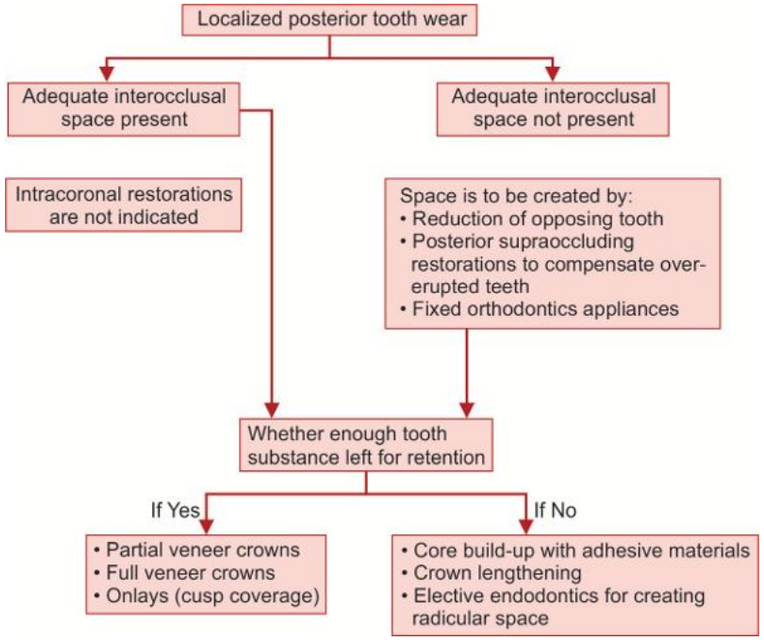
Flowchart 26.2: Management of generalized tooth wear



Flowchart 26.3: Management of localized anterior tooth wear



Flowchart 26.4: Management of localized posterior tooth wear



The management of generalized tooth wear, localized anterior tooth wear and localized posterior tooth wear is summarized in Flowcharts 26.2, 26.3 and 26.4, respectively.

C. Treatment of Attrition

The management of attrition includes:

- Physiological attrition requires no treatment because formation of secondary dentin, tooth eruption and alveolar bone growth compensate for the occlusal attrition. Cementum may also be associated as a compensatory mechanism.
- When the attrition becomes pathological, where it progresses fast, it will lead to malocclusion, difficulty

in mastication, pain in TMJ and/or total collapse of occlusion. In such instances, splint therapy may be indicated for the remaining dentition.

- The treatment of severe attrition, where teeth are worn to the gingival margins, may require restoration of the vertical dimension to improve function and esthetics.

Treatment varies, depends upon the following categories:

Category I : Appearance is satisfactory.
Category II : Appearance is unsatisfactory and there is no need to raise the vertical height.
Category III : Appearance is unsatisfactory, but there is need to raise the vertical height, which in turn depends on the availability of space, whether it is present or needs to be created.

A. *Category I* (Appearance is satisfactory)

a. *Counseling:* Counseling is required in patients with parafunctional habits. Habit breaking appliance should be given in patients with bruxism or clenching.

b. *Conventional restorative treatment*

- Exposed pits are filled (Fig. 26.23A, B).
- Occlusal disharmony is corrected.
- Consider crown lengthening procedure (Fig. 26.24)

Monitoring of such cases can be carried out by taking silicone rubber impressions every three months and evaluating the casts under microscope.

B. *Category II* (Appearance is unsatisfactory, but there is no need to raise the vertical height)

- Teeth are restored, preferably with all ceramic crowns. The crowns can manage occlusal attrition as well as fractured cusps (Fig. 26.25A, B).
- Occlusal guard for protection against nocturnal clenching, like bleaching trays, etc.



Fig. 26.23A: Incisal surface filled with glass-ionomer cement



Fig. 26.23B: Occlusal pits filled with glass-ionomer cements



Fig. 26.24: Crown lengthening



Fig. 26.25A: Tooth preparation for full crowns

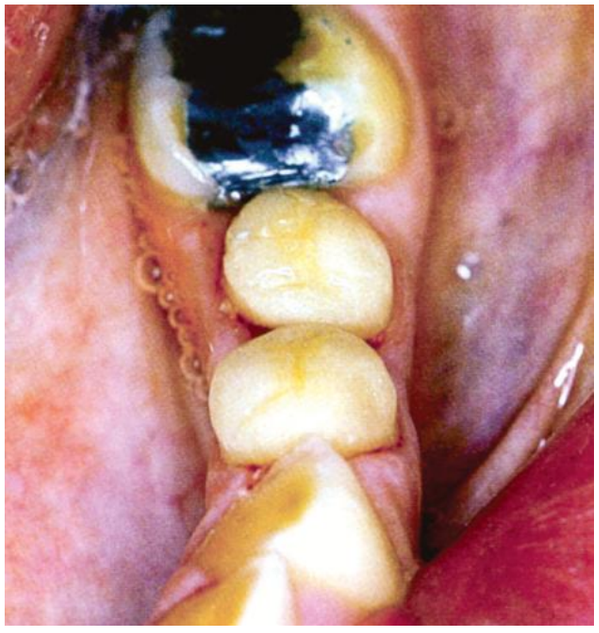


Fig. 26.25B: Crowns restored

C. *Category III* (Appearance is unsatisfactory and there is a need to raise the vertical height)

- Generalized increase in vertical height is required [vertical dimensions of posterior tooth is increased, as per requirement and as conducive to temporomandibular joint (Fig. 26.26A, B)].
- Orthodontic tooth movement can be used for over-eruption of posterior teeth, creating space for the anterior teeth; the created space is to be utilized in retruded cusp position (RCP) and intercuspal position (ICP).

Points to remember

- No occlusal cutting; only spicules need to be removed or filled.
- No disturbance of interdental col area.
- Retention of the prosthesis is achieved by surface area.
- Buccal and lingual reduction takes care of periodontal health.

Dahl Concept

It is a method of gaining space in case of localized tooth wear, where there is insufficient space available in either centric occlusion or centric relation. It refers to the relative axial movement that occurs when a localized restoration is placed in supraocclusion and the occlusion re-establishes its full arch contacts. Dahl and Krungstad reported that interocclusal space would be created through a process of combined intrusion (40%) and extrusion (60%). Most of the cases re-establish occlusal contacts within 4–6 months; however, it may take 18–24 months.



Fig. 26.26A: Raising vertical height, showing one side

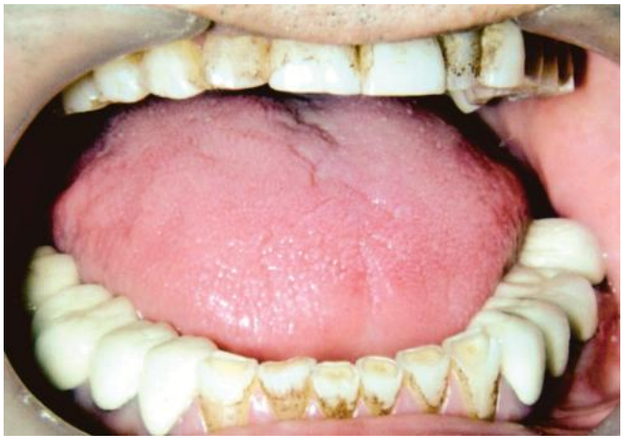


Fig. 26.26B: Raising vertical height, showing both sides

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